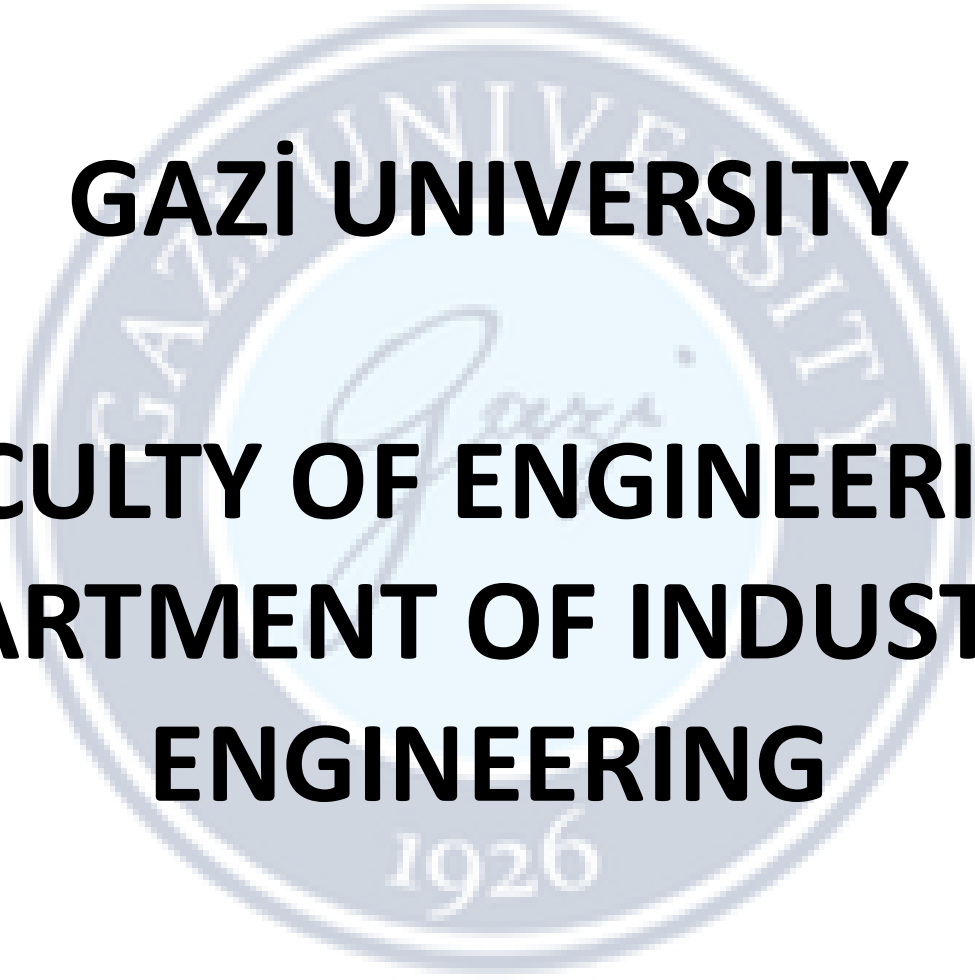


The logo of Gazi University is a circular seal. It features the university's name in Turkish, 'GAZİ ÜNİVERSİTESİ', around the top inner edge and the year '1926' at the bottom. In the center is a stylized signature of 'Gazi'.

GAZİ UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

IE104-COMPUTER PROGRAMMING I

WEEK 7: CONDITIONAL STATEMENTS

Conditional Statements

- In some cases, it may be necessary to change the flow of the program or perform different operations. Sometimes making choices or repeating actions are needs for a program in a certain number of times.
- Condition statements and loop structures are used to perform such operations.
- Two different statements can be used to perform conditional operations in C#:
 - if statement
 - switch statement

IF Statement

- The main statement of flow controls .
- Allows different operations according to the condition.
- Simple to use:

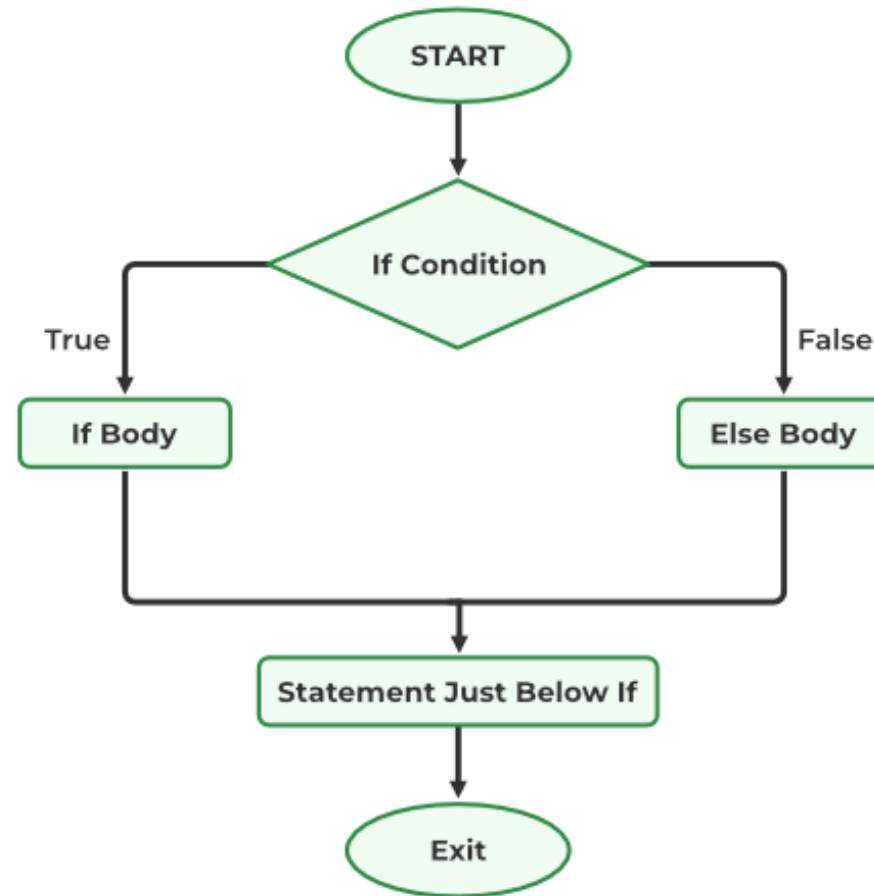
```
if (condition)
{
    // execute code statements of if block when
    // condition is true
}
else
{
    // execute code statements of if block when
    // condition is true
}
```

OR

```
if (condition 1)
{
    // execute code statements of if block when
    // condition is true
}
else if (condition2)
{
    // Code block executed if condition1 is false and
    // condition2 is true
}
else
{
    // Code block executed if both condition1 and condition2
    // are false
}
```

- If there is more than 1 line in the statement region, it is mandatory to use Curly Brackets ({ }) in a program block.

IF Flow Chart



Example program for IF statement

```
using System;

public class NumberChecker
{
    public static void Main(string[] args)
    {
        // Let's assume the number is already defined here
        int number = 42;

        // Core if-else if-else logic
        if (number > 0)
        {
            Console.WriteLine("The number is positive.");
        }
        else if (number < 0)
        {
            Console.WriteLine("The number is negative.");
        }
        else
        {
            Console.WriteLine("The number is zero.");
        }
    }
}
```

Work on your own

```
if (condition 1)
{
    // execute code statements of if block when
    // condition is true
}

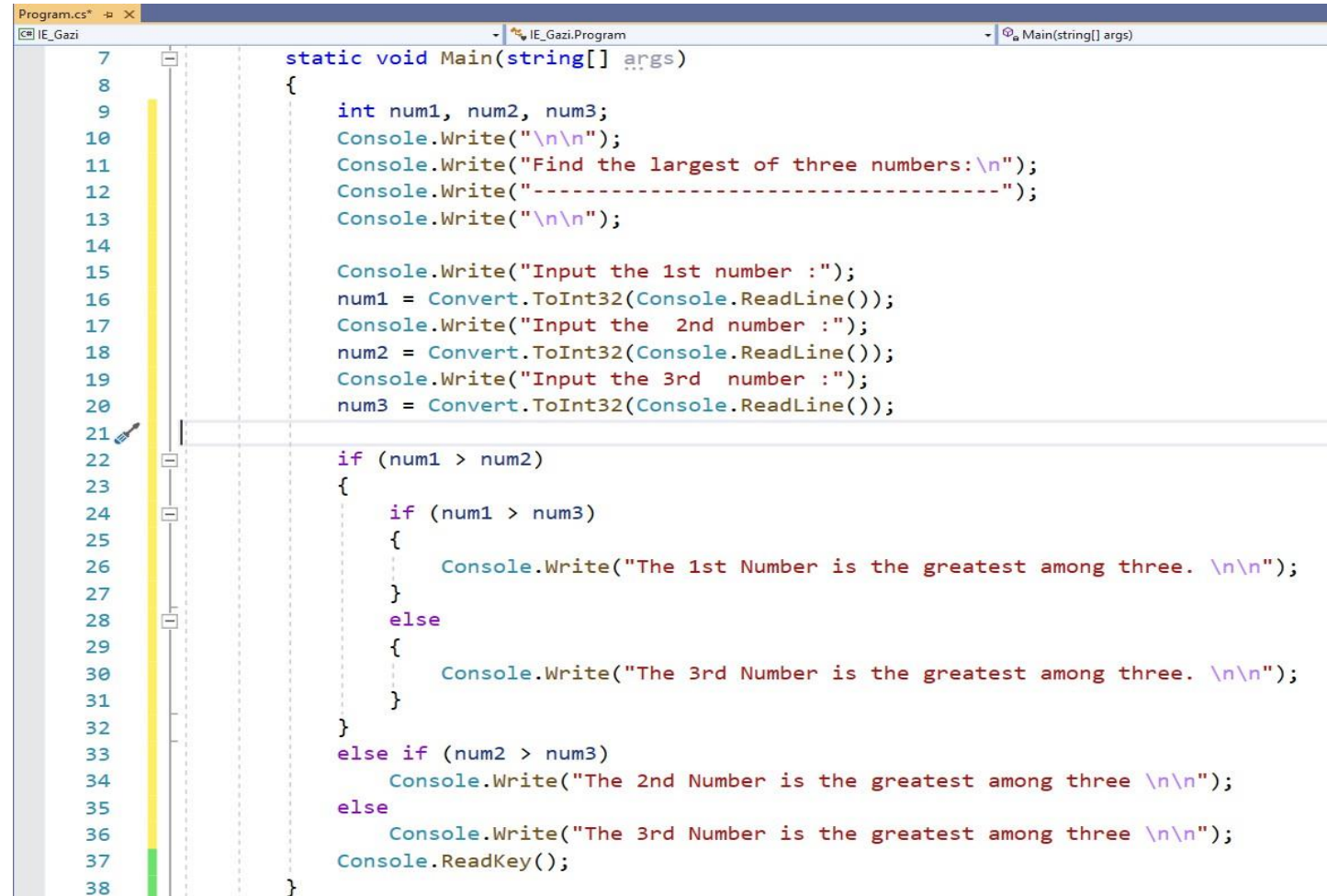
else if (condition2)
{
    // Code block executed if condition1 is false and
    // condition2 is true
}

else
{
    // Code block executed if both condition1 and condition2
    // are false
}
```

Given else if statement, Now try to print if number is exactly zero.

IF Statement

Write a C# program to find the largest number among 3 numbers which are entered through keyboard as inputs.



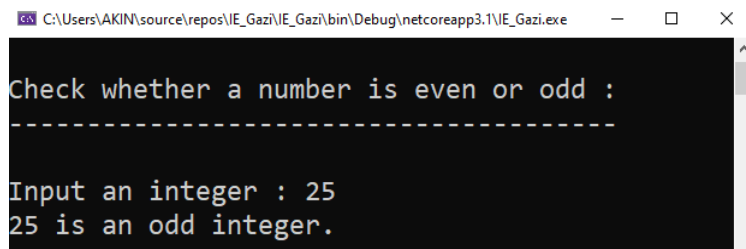
```
Program.cs* - X
IE_Gazi
IE_Gazi.Program
Main(string[] args)

7      static void Main(string[] args)
8      {
9          int num1, num2, num3;
10         Console.WriteLine("\n\n");
11         Console.WriteLine("Find the largest of three numbers:\n");
12         Console.WriteLine("-----");
13         Console.WriteLine("\n\n");
14
15         Console.WriteLine("Input the 1st number :");
16         num1 = Convert.ToInt32(Console.ReadLine());
17         Console.WriteLine("Input the 2nd number :");
18         num2 = Convert.ToInt32(Console.ReadLine());
19         Console.WriteLine("Input the 3rd number :");
20         num3 = Convert.ToInt32(Console.ReadLine());
21
22         if (num1 > num2)
23         {
24             if (num1 > num3)
25             {
26                 Console.WriteLine("The 1st Number is the greatest among three. \n\n");
27             }
28             else
29             {
30                 Console.WriteLine("The 3rd Number is the greatest among three. \n\n");
31             }
32         }
33         else if (num2 > num3)
34             Console.WriteLine("The 2nd Number is the greatest among three \n\n");
35         else
36             Console.WriteLine("The 3rd Number is the greatest among three \n\n");
37         Console.ReadKey();
38     }
```

IF Statement

Write a C# program to check whether a given number is even or odd.

Expected screen has shown below:



```
C:\Users\AKIN\source\repos\IE_Gazi\IE_Gazi\bin\Debug\netcoreapp3.1\IE_Gazi.exe
Check whether a number is even or odd :
-----
Input an integer : 25
25 is an odd integer.
```

```
0 başvuru
static void Main(string[] args)
{
    // Example for Gazi IE
    int num1, rem1;
    Console.Write("\n\n");
    Console.Write("Check whether a number is even or odd :\n");
    Console.Write("-----");
    Console.Write("\n\n");
    Console.Write("Input an integer : ");
    num1 = Convert.ToInt32(Console.ReadLine());
    rem1 = num1 % 2;
    if (rem1 == 0)
        Console.WriteLine("{0} is an even integer.\n", num1);
    else
        Console.WriteLine("{0} is an odd integer.\n", num1);

    Console.ReadKey();
}
```

IF Statement

Writing a simple calculator program in C#

CONTINUE TO WRITE HERE

```
static void Main(string[] args)
{
    Console.WriteLine("Enter the action to be performed");
    Console.WriteLine("Press 1 for Addition");
    Console.WriteLine("Press 2 for Subtraction");
    Console.WriteLine("Press 3 for Multiplication");
    Console.WriteLine("Press 4 for Division");
    Console.WriteLine("Press 5 for Mod Operation \n");
    Console.Write("What is your choice?:");

    int action = Convert.ToInt32(Console.ReadLine());
    int input_1=0, input_2=0 ;
    if (action>0&&action<6)
    {
        Console.Write("Enter 1st input:");
        input_1 = Convert.ToInt32(Console.ReadLine());
        Console.Write("Enter 2nd input:");
        input_2 = Convert.ToInt32(Console.ReadLine());
    }
    else
        Console.WriteLine("Wrong choice. Try again later.");
```

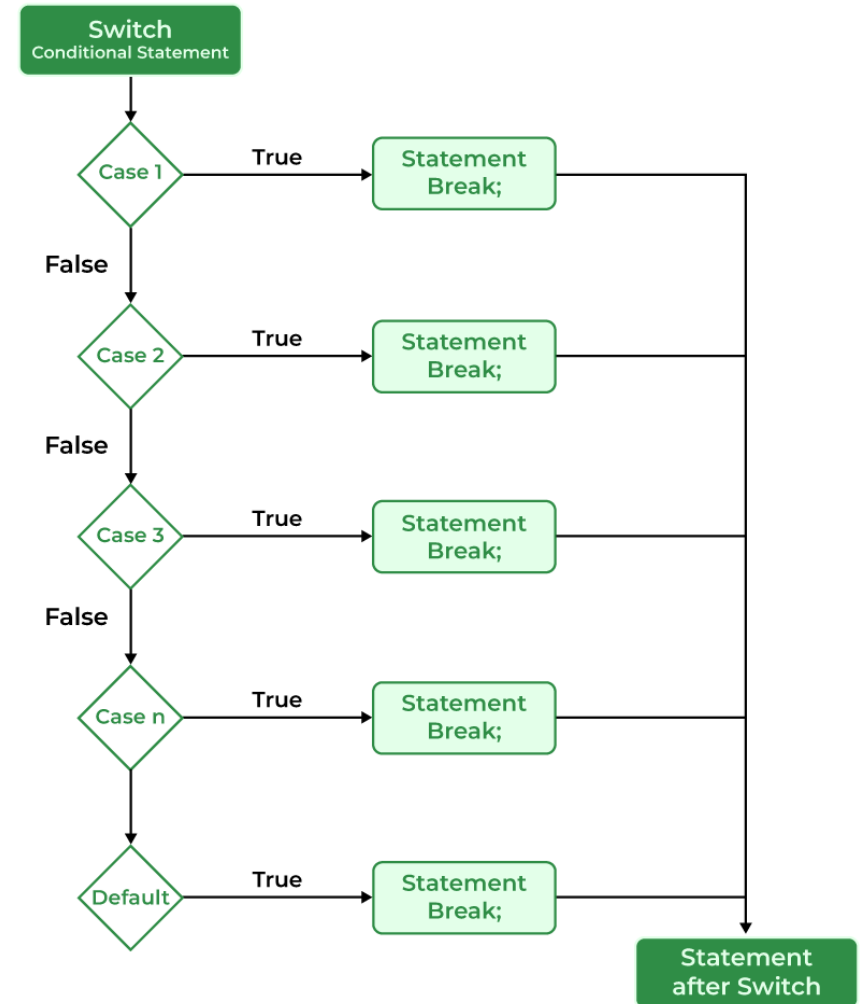


```
if (action==1)
    Console.WriteLine("Addition Result={0}", input_1 + input_2);
else if (action == 2)
    Console.WriteLine("Subtraction Result={0}", input_1 - input_2);
else if (action == 3)
    Console.WriteLine("Multiplication Result={0}", input_1 * input_2);
else if (action == 4)
{
    if(input_2==0)
        Console.WriteLine("Divider cannot be zero.");
    else
        Console.WriteLine("Division Result={0}", (float)input_1 / (float)input_2);
}
else if (action == 5)
{
    if (input_2 == 0)
        Console.WriteLine("Divider cannot be zero.");
    else
        Console.WriteLine("The remainder of division={0}", input_1 % input_2);
}
```

Switch-Case Statement

- Used to perform operations due to the different values that an expression can take.
- It can also make if statements simpler.

```
switch(variable)
{
    case 1:
        //execute your code
        break;
    case n:
        //execute your code
        break;
    default:
        //execute your code
        break;
}
```



Switch-Case Statement

- You must terminate each case statement with a skip (**break, goto, return, continue**) statement.
- The **Continue** statement can be used if the switch statement is in the body of a loop.
- A skip statement is not required if no statement is given in a case
 - Case 2: // no statement
 - Case 3:
 - Case 4: expression....;break,;
 - Case 5: expression...; goto case 4;

Switch-Case Statement

There are some rules to be aware of when using a switch statement:

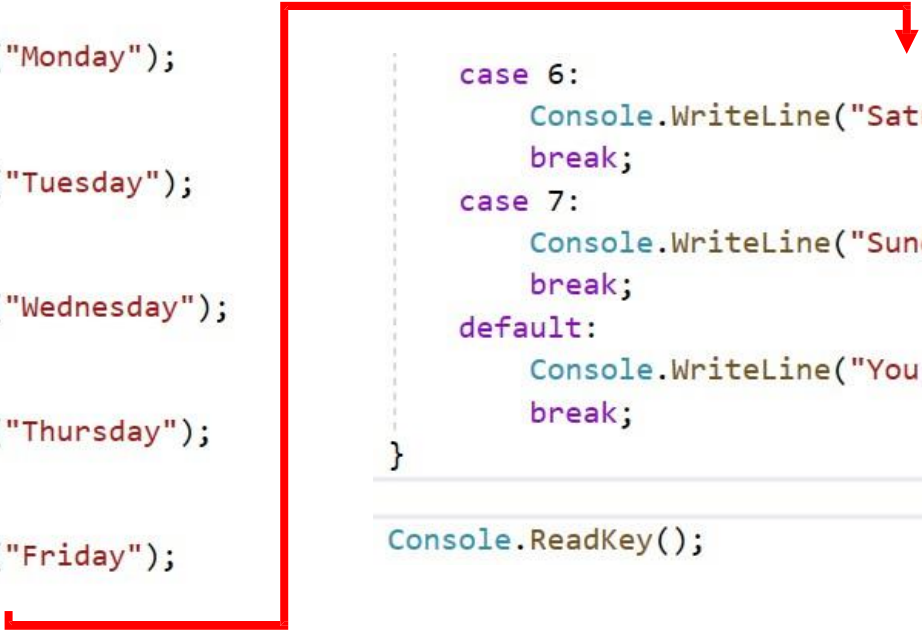
- Statements after the “**case**” word must be constants.
- “**case**” expressions can be integer, character, or string constants.
- “**default**” and “**case**” statements can be written in any order.
- There cannot be more than one same “**case**” statement in the same switch block.
- **It is not** mandatory to use “**default**”.
- To redirect the flow from one “**case**” statement to another “**case**” statement, the “**goto**” keyword is used.

Switch-Case Statement

```
static void Main(string[] args)
{
    int day;

    Console.Write("Which weekday would you like to be in?:");
    day = int.Parse(Console.ReadLine());
    switch (day)
    {
        case 1:
            Console.WriteLine("Monday");
            break;
        case 2:
            Console.WriteLine("Tuesday");
            break;
        case 3:
            Console.WriteLine("Wednesday");
            break;
        case 4:
            Console.WriteLine("Thursday");
            break;
        case 5:
            Console.WriteLine("Friday");
            break;
        case 6:
            Console.WriteLine("Saturday");
            break;
        case 7:
            Console.WriteLine("Sunday");
            break;
        default:
            Console.WriteLine("You are not from this planet.");
            break;
    }

    Console.ReadKey();
}
```



Switch-Case Statement

```
class Program
{
    //Switch -Case Example for GaziIE
    0 başvuru
    static void Main(string[] args)
    {
        string myinput;
        int myint;

        begin:

        Console.WriteLine("Pelase enter a number between 1-3 (inc.1-3):");
        myinput = Console.ReadLine();
        myint = int.Parse(myinput);

        switch (myint)
        {
            case 1:
                Console.WriteLine("You have entered 1 as an input");
                break;
            case 2:
                Console.WriteLine("You have entered 2 as an input");
                break;
            case 3:
                Console.WriteLine("You have entered 3 as an input");
                break;
            default:
                Console.WriteLine("Your number is not between 1-3");
                break;
        }
    }
}
```

decide:

```
Console.WriteLine("\nType \"continue\" to go on or \"quit\" to stop:");
myinput = Console.ReadLine();

switch (myinput)
{
    case "continue":
        goto begin;
    case "quit":
        Console.WriteLine("GoodBye!");
        break;
    default:
        Console.WriteLine("Incorrect input");
        goto decide;
}
```

THE END



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GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM